

UC LINA 600 LINEAR ALGEBRA FOR GRAPHICS · MODULE 03 · LECTURE NOTES

Matrices as Linear Maps

The columns are where the basis vectors land.

MODULE OUTCOMES

- Read a matrix as a transformation.
- Explain why the columns are the images of basis vectors.
- Explain matrix multiplication as composition.
- Explain why multiplication is not commutative.
- Build the standard graphics transform matrices.

1 The one idea

THE COLUMNS OF A MATRIX ARE WHERE THE BASIS VECTORS LAND — AND THAT SINGLE FACT DERIVES EVERYTHING

A linear map is completely determined by what it does to the basis vectors, because every other vector is a linear combination of them and the map preserves linear combinations (§4's definition) — so there is nothing left to specify.

So write down where each basis vector goes, stack those images as columns, and you have the matrix — which is how you build every transform in this course, rather than looking one up and hoping the sign convention matches yours.

And then Mv means 'take v 's coordinates as weights, and combine the columns accordingly' — which is exactly what matrix-vector multiplication computes, read correctly rather than as a row-by-column ritual.

Which makes the definition of the product inevitable rather than arbitrary — and anybody who finds matrix multiplication unmotivated has been shown the rule before the reason, which is unfortunately the standard order.

WHERE DO THE BASIS VECTORS GO under a rotation by theta in the plane?

$$\begin{aligned}(1,0) &\rightarrow (\cos t, \sin t) \\ (0,1) &\rightarrow (-\sin t, \cos t)\end{aligned}$$

Those are the columns. So

$$R = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

AND THAT IS THE DERIVATION. No memorisation, and no risk of getting the signs backwards -- draw the two arrows and read the answer off the picture.

SAME METHOD, SCALE BY (a, b):

$$\begin{aligned}(1,0) &\rightarrow (a, 0), \quad (0,1) \rightarrow (0, b) \\ S &= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}\end{aligned}$$

SAME METHOD, SHEAR:

$$(1,0) \rightarrow (1, 0), \quad (0,1) \rightarrow (k, 1)$$

and you have it.

EVERY MATRIX IN THIS COURSE IS BUILT THIS WAY.

Draw where the basis vectors go and read off the columns — which removes every sign error from every transform in the course, and is why Project 1 requires every matrix built this way rather than looked up. The ten seconds spent drawing two arrows is reliably cheaper than the twenty minutes spent wondering why the object rotates the wrong way.

2 Composition

MATRIX MULTIPLICATION IS COMPOSITION OF MAPS, WHICH IS WHY IT IS ASSOCIATIVE AND WHY IT IS NOT COMMUTATIVE

AB means 'apply B first, then apply A' — right to left — which is backwards from reading order and is the source of endless confusion about transform order, including in documentation.

It is associative because composing functions is associative: (AB)C and A(BC) both mean 'do C, then B, then A', which is obviously the same thing once you read the product as composition — and is a tedious computation if you read it as entries.

And it is not commutative because the order in which you apply operations matters — rotate then translate lands an object somewhere quite different from translate then rotate — which you can verify with a pen and a piece of paper in ten seconds.

So the non-commutativity is not an algebraic quirk to be remembered — it is a geometric fact that the algebra faithfully records — and Project 1 asks you to demonstrate it and explain it rather than merely observe it.

3 The standard transforms

Transform	What it does to the basis vectors	Notes
Scale	Each basis vector stretched along its own axis.	A diagonal matrix.
Rotation	Both basis vectors turned by the same angle.	Columns are cosines and sines — derive them, do not recall them (§1).
Shear	One fixed, the other tilted.	A single off-diagonal entry.
Reflection	One basis vector negated.	Determinant -1 , which flips handedness (Module 09 §3).
Projection onto a subspace	Collapsed onto a line or plane.	Not invertible — information is destroyed, and the determinant is zero.
Translation	— it is not a linear map at all.	Needs Module 07's fourth coordinate — see the note.

Translation is not a linear map — it moves the origin, and every linear map necessarily fixes the origin (set $c = 0$ in $T(cv) = cT(v)$) — which is precisely why homogeneous coordinates exist (Module 07). The translation row is the one that motivates the whole of Module 07, and noticing the gap here makes that module feel like a solution rather than a convention.

4 What linear means

- $T(u + v) = T(u) + T(v)$ and $T(cv) = c \cdot T(v)$ — which together mean the map preserves linear combinations, and that is the entire definition; everything else in this module is a consequence.
- Geometrically: straight lines stay straight, the origin stays where it is, and parallel evenly-spaced lines stay parallel and evenly spaced — which is the picture to check a candidate map against.
- Which immediately rules out translation (§3) — it moves the origin — and rules out anything involving a squared term, a sine of a coordinate, or a constant offset.
- And it is exactly why a linear map is determined by its action on a basis (§1): knowing the images of the basis vectors plus linearity determines the image of everything else, by writing it as a combination.
- So the restriction buys the representation — linear maps are precisely the maps a matrix can express — which is the trade, and is why the definition is worth being exact about rather than treating 'linear' as a vague synonym for simple.

AND THE THING TO CARRY FORWARD

Never look up a transform matrix — draw where the basis vectors go and read off the columns (§1) — which takes ten seconds and cannot be got backwards, unlike a half-remembered formula.

And always read a matrix product right to left, as composition — which makes the model-view-projection chain readable as a sentence (Module 08's Viewport · Project · View · Model is 'model first, viewport last').

Plus: translation is coming, and it does not fit into this framework — which is the one gap in this module and is Module 07's entire subject.

Which is a good example of how this course is built: a restriction that buys you something, and then a trick that evades the restriction without giving the purchase back — homogeneous coordinates let you express translation as a linear map in one more dimension.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
3Blue1Brown, chapters 3 to 5 (free)	§§1 and 2 — matrices as transformations and multiplication as composition, which is this module's entire thesis.
MIT 18.06, lectures 2 and 3 (free video)	§§1 and 4 — the column picture, which Strang builds the course on.
Immersive Math, chapter 6 (free)	§§1 to 3 — drag a transform's basis images and watch the matrix change, which is §1 made interactive.
Lengyel, chapters 2 and 3 (library copy)	§3 — the graphics transforms in the conventions you will meet in an engine.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive the 2D rotation matrix** by drawing the basis images.
- 2 **Derive scale, shear, and reflection** the same way.
- 3 **Derive a 3D rotation about each axis.**
- 4 **Verify Mv combines the columns**, on three examples.
- 5 **Compose two transforms in both orders** and draw the results.
- 6 **Explain the difference geometrically**, not just numerically.

- 7 **Verify associativity** on a triple product.
- 8 **Show that translation violates linearity**, explicitly.
- 9 **Find a projection matrix** onto a line and show it is not invertible.
- 10 **Check the two linearity conditions** for five candidate maps.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What are the columns of a matrix?
- 2 Why is a linear map determined by the basis?
- 3 What does Mv compute, read correctly?
- 4 Derive the rotation matrix.
- 5 What does AB mean, and in what order?
- 6 Why is multiplication associative? Why not commutative?
- 7 Name six standard transforms and their effect on the basis.
- 8 Why is translation not linear?
- 9 State the two linearity conditions and their geometric meaning.
- 10 What does the restriction buy?